Two-Step Diffusion for KPP Equations: Sections 4.3 and 4.4

KPP Code Reproduction

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1 KPP Front Speed Experiment on T^2

This section follows the reference paper's Section 4.3. The goal is to estimate the front speed of the KPP equation in a two-dimensional, time-periodic cellular flow and to generate particle clouds that resemble the long-horizon invariant measure of the underlying Feynman–Kac particle system.

1.1 Problem Setting and Feynman–Kac Particle System

The reaction–diffusion–advection equation is

$$u_t + v(x) \cdot \nabla u = \kappa \Delta u + ru(1-u). \tag{1}$$

Here κ is the diffusion constant, v is the prescribed velocity field, and u is the reactant concentration. The minimal front speed along a direction e is written as

$$c^*(e) = \inf_{\lambda > 0} \frac{\mu(\lambda)}{\lambda}, \tag{2}$$

where $\mu(\lambda)$ is the principal eigenvalue of a twisted parabolic operator. For a fixed λ, the corresponding twisted diffusion on $T^2 = [0, 2\pi)^2$ is

$$dX_t = (2\kappa\lambda e_1 + v(X_t, t))\, dt + \sqrt{2\kappa}\, dW_t. \tag{3}$$

The cellular velocity field is

$$v(x,t) = (-\cos x_2 - \theta \cos(2\pi t) \sin x_1,\ \cos x_1 + \theta \cos(2\pi t) \sin x_2)\,. \tag{4}$$

The Feynman–Kac weights use the potential

$$V_\lambda(x,t) = \kappa\lambda^2 + \lambda v(x,t) \cdot e_1. \tag{5}$$

This particle representation provides the reference invariant measures used for training and evaluation.

1.2 Experimental Setup

The 2D experiment uses the parameters listed in Table 1.

Three initialization modes are compared: cold start, MeanFlow warm start, and MeanFlow plus DP warm start. Two evaluation metrics are used. The first is eigenvalue convergence of $\hat{c}_T(\lambda)$ for $T = 1, 2, 4, \ldots$. The second is the invariant-measure W_2 on the torus. In practice, the invariant-measure error is evaluated with a 72×72 histogram and the periodic ground metric.

Table 1: 2D KPP experimental parameters.

Parameter	Value
Domain	$T^2 = [0, 2\pi)^2$
λ	2.0
θ	1.0
Time step dt	2^{-8}
Substeps per period r_n	256
Particles per level M	2×10^4
Long-horizon generations T	4096
Cold start	uniform on T^2
Training $\log_2 \sigma$	$-2.00, \ldots, -3.75$
Evaluation $\log_2 \sigma$	$-2.00, \ldots, -4.50$

1.3 Preliminary Results

Table 2 reports current W_2 values. The two-stage pipeline is end-to-end functional. The DP stage reduces the error in several diffusion regimes, but the current MeanFlow baseline is still higher than the reference paper baseline.

Table 2: 2D KPP preliminary W_2 results.

$-\log_2 \sigma$	cold	MeanFlow	MF+DP
2.00	1.054	0.256	0.123
2.50	1.090	0.234	0.112
3.00	1.114	0.221	0.125
4.00	1.150	0.238	0.195
4.50	1.172	0.236	0.261

Update interface. Replace these rows with the final `metrics.json` values after the current 2D training completes.

2 3D KPP Front Speed Experiment with Time-Dependent Kolmogorov Flow

This section follows the reference paper's Section 4.4. The experiment extends the two-stage approach to a three-dimensional, time-dependent Kolmogorov flow.

2.1 Problem Setting

The governing equation is the same KPP equation, but the spatial domain is $T^3 = [0, 2\pi)^3$. The velocity field is

$$v(x, t) = (\sin(x_3 + \sin(2\pi t)), \sin(x_1 + \sin(2\pi t)), \sin(x_2 + \sin(2\pi t))) \,. \tag{6}$$

This flow is divergence-free and generates more complex anisotropic invariant measures than the 2D cellular flow.

2.2 Experimental Setup

Table 3 summarizes the 3D parameters. The discretization is the same as in the 2D experiment.

Table 3: 3D KPP experimental parameters.

Parameter	Value
Domain	$T^3 = [0, 2\pi)^3$
λ	2.0
Time step dt	2^{-8}
Substeps per period r_n	256
Particles per level M	2×10^4
Long-horizon generations T	4096
Training $\log_2 \sigma$	$-2.00, \ldots, -3.75$
Evaluation $\log_2 \sigma$	$-2.00, \ldots, -4.50$

The invariant-measure error is computed on the x-z projection:

$$E_{xz}(\sigma) = W_2\left((\Pi_{xz})_{\#}\hat{\mu}(\sigma), (\Pi_{xz})_{\#}\hat{\nu}(\sigma)\right), \tag{7}$$

where $\Pi_{xz}(x, y, z) = (x, z)$. Eigenvalue convergence is also reported for warm and cold initializations.

2.3 Preliminary Results

Table 4 reports representative current W_2 values. The 3D MeanFlow baseline is closer to the paper than the 2D baseline, but the DP improvement remains modest.

Table 4: 3D KPP preliminary W_2 results.

$-\log_2 \sigma$	cold	MeanFlow	MF+DP
2.00	0.889	0.151	0.150
2.50	0.933	0.136	0.135
3.00	0.922	0.131	0.129
4.00	0.957	0.133	0.132
4.50	0.956	0.153	0.153

Update interface. After full 3D training, replace these values and record the actual seed from `actual_seed.json`.